

IVAN YANG

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Technical Skills

Languages/Tools: Golang · gRPC · C# · TypeScript · Redis · Python · Git · Docker · Kubernetes

AWS Services: Lambda · RDS · DynamoDB · S3 · ECS · SQS · SNS · ELB · EventBridge

Other Tech Stacks: PostgreSQL · WebSocket · REST API · MongoDB · K6 · Cloudflare R2

Certifications

AWS Certified Solutions Architect – Associate (SAA-C03)

Certificate in Artificial Intelligence, University of Toronto School of Continuing Studies

Experience

IGG Canada

Oct 2023 - Feb 2026

Software Engineer (Generalist)

Vancouver, Canada

- Developed in a distributed game server backend in **Golang** and complex UI systems in **C#**, serving gameplay features of large-scale mobile games *Mythic Heroes* and *Last Trigger*, supporting thousands of concurrent real-time players.
- Set up multi-environment GitLab CI/CD pipelines with Docker, supporting start/stop/restart/backup/restore operations across multiple production clusters; maintained emergency hotfix mechanisms for zero-downtime patches.
- Led UI/UX optimization groups by maintaining component libraries of common widgets, ensuring visual consistency across games and standardizing UI patterns using modern frontend best practices.
- Designed configuration frameworks to empower game designers in integrating and adjusting configurations without requiring code changes.
- Improved game performance by optimizing tick-level operations within hotpaths and replacing redundant game objects with dynamic instantiation, leading to an 80% reduction in draw calls. Enhanced code structure and maintainability by refactoring state controls into FSMs.
- Implemented and maintained data centers to parse server data and developed APIs for other in-game features.
- Wrote unit tests for server APIs and actively contributed to end-to-end testing in preparation for game releases.

Centivizer, MIE Interactive Media Lab

May 2022 – Sept 2022

React Developer

Toronto, Canada

- Developed *Emotional Mole*, a browser game designed to measure long-term care residents' reaction time.
- Refined the backend API by optimizing routing in Express.js and designing new MySQL schema with referential integrity; fetching time was reduced from 120ms to 75ms; received positive feedback from end users.
- Implemented a user authentication system for game records, and created ranking leaderboards for various game modes.

Projects

Snap2Send | Go, TypeScript, React Native (Expo), Electron, Cloudflare R2, Neon Postgres

- Built a cross-platform notebook and file-sharing app shipping to iOS, Android, web, and desktop from a single React Native + Expo codebase wrapped in Electron for the desktop target.
- Designed a Go REST backend on `net/http` + `pgx/v5` backed by Neon Postgres, exposing endpoints for rooms, messages, file transfers, and authentication.
- Implemented multipart streaming uploads to Cloudflare R2 via the AWS S3 SDK, with a transfer table tracking upload state and a backend-proxied download route to keep the R2 bucket private.
- Built a server-side image-processing pipeline that center-crops and resizes user-uploaded avatars to 256×256 JPEG before storage, served via a public R2 prefix with cache-busting query strings.
- Integrated OAuth 2.0 across multiple providers with three parallel flows: full-page redirect on web, deep-link callback on iOS/Android, and intercepted BrowserWindow protocol handler in Electron.
- Designed a single session model unifying OAuth and email/password auth, using opaque 64-character DB-backed tokens with a 30-day TTL and bcrypt-hashed credentials.
- Implemented a four-layer abuse defense combining Cloudflare edge WAF, per-IP token-bucket rate limiting, per-user daily quotas with lazy UTC-day rollover, and in-memory sliding-window login lockout, all backed by a Postgres audit log.
- Deployed the Go backend to AWS ECS Fargate with task definitions sourcing secrets from SSM Parameter Store, fronted by Cloudflare for edge caching, WAF, and TLS; hosted the web build on Cloudflare Pages with `_redirects`-based friendly URLs routing to platform-specific app store pages.

Education

University of Toronto, St. George

Sept 2018 – June 2023

Honours Bachelor of Science in Statistics, Computer Science, and Mathematics

Toronto, Canada

Graduate Coursework: Generative Modeling Research, Computer Vision, Deep Learning & Neural Nets, Machine Learning

Scholarship Recipient, University of Toronto President's Scholarship Program, 2018

Activities: Recognized Group Leader, *Enriched Data Structures and Analysis*